

$$MSE = \frac{1}{n} \sum_{i=0}^n \underbrace{(y - y')^2}_{L_1} \times \boxed{h=5}$$

$$\begin{array}{l} 7 - 6 = 1 \\ 5 - 4 = 1 \\ 8 - 11 = 3 \\ 4 - 6 = 2 \\ 9 - 8 = 1 \end{array} \left. \begin{array}{l} 1 \\ 1 \\ 9 \\ 4 \\ 1 \end{array} \right\} \begin{array}{l} L_1 \\ L_2 \end{array} \quad (1 + 1 + 9 + 4 + 1) = 16 \quad \sqrt{L_2}$$

$$\frac{1}{5} (16) = \frac{16}{5} = \boxed{3.2} \quad \checkmark$$

$\sqrt{L_1} = 8 \quad \checkmark$

$$f(x) \rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 10 \end{pmatrix}, \begin{pmatrix} 2 \\ 7 \end{pmatrix} \quad m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{7 - 2}{10 - 0} = \frac{5}{10} = \boxed{0.5}$$

$$\boxed{y = 0.5x + 2}$$

y	y'	L ₁	L ₂
2.5	2.5	0	0
3	3	0	0
3.5	3.5	0	0
* 6	4	2 ✓	4 ✓
4.5	4.5	0	0
5	5	0	0
5.5	5.5	0	0
6	6	2 ✓	4 ✓
* 6.5	6.5	0	0
7	7	0	0

$$\boxed{L_1 = 4}$$

$$\boxed{\Sigma}$$

$$\boxed{L_2 = 8} \quad \checkmark$$

$$\frac{1}{n} \Sigma (L_2) \quad n = 10$$

$$\sqrt{ECM} = \frac{1}{10} (8) = \boxed{0.8}$$

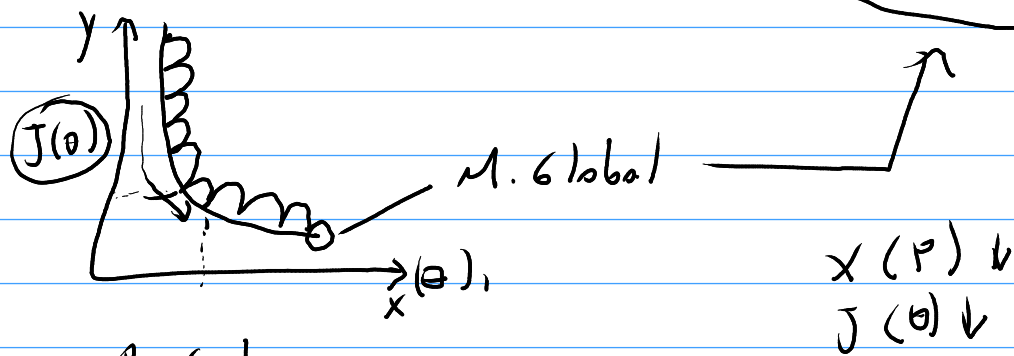
$$y' = \underbrace{w}_{H} x_1 + b \quad \checkmark$$

$$\boxed{y' = 0.03 x_1 + 16} \quad \checkmark$$

$$\boxed{ECM = 11.03} \quad \checkmark$$

w, b

1. Gradiente $w_b \rightarrow \sum w_{b,activation} \rightarrow \boxed{\text{Pérdida}} \downarrow \text{min}$



R. Cadena

$$J(x) = (x-3)^2 \rightarrow J'(x)$$

$$x^{1-1} = x^0 = 1$$

$$* J'(x) = 2(x-3)(1)$$

$$\checkmark x=10 \quad \checkmark x=5 \quad x=3 \quad x=0$$

$$J'(10) = 2(10-3)(1) = 2(7) = 14 \downarrow$$

$$J'(5) = 2(5-3)(1) = 2(2) = 4 \downarrow$$

$$J'(3) = 2(3-3)(1) = 0 \quad \checkmark$$

$$J'(1) = 2(1-3)(1) = 2(-2) = -4$$

$$J'(0) = 2(0-3)(1) = 2(-3) = -6$$

$$\begin{array}{l} 2x^2 \\ 2 \cdot 2 \cdot x^{2-1} \\ 4x \end{array}$$

T. Aprendizaje $\checkmark$

$$\eta - \alpha \uparrow \downarrow \leftarrow \begin{array}{l} w \\ b \end{array}$$

$$f(x) = x - J'(x)$$

$$f(x) = x - \frac{1}{2} J'(x)$$

$$\boxed{\alpha = \frac{1}{2}}$$

$$\rightarrow f(x) = x - \frac{1}{2} 2(x-3)$$

$$[x=10 \quad x=5 \quad x=3]$$

$$f(10) = 10 - \frac{1}{2} (2(10-3)) = 10 - 7 = 3$$

$$f(5) = 5 - \frac{1}{2} (2(5-3)) = 5 - 2 = 3$$

$$f(3) = 3 - \frac{1}{2} (2(3-3)) = 3$$

D. Gradiente *

$$\boxed{\varepsilon_{CM} = L}$$

① $w_v = 0 \quad b_v = 0 \quad \eta = 0.05$

$$X = [1-4]$$

$$\sqrt{1, 2, 3, 4}$$

$$\boxed{n = 4}$$

$$\boxed{y = 0.5x + 2}$$

$$w \rightarrow \frac{\partial L}{\partial w} = -\frac{2}{n} \sum_{i=1}^n (y_i - y_i') x_i$$

$y_i^{(0)} = 0$

x	$y_i(0.5x+2)$	L_2
1	$0.5(1)+2 = 2.5$	6.25
2	$0.5(2)+2 = 3$	9
3	$0.5(3)+2 = 3.5$	12.25
4	$0.5(4)+2 = 4$	16

$$\varepsilon_{CM} = \frac{43.5}{4} = \boxed{10.875}$$

$$\boxed{L_2 = 43.5}$$

$$\sum (2.5 + 6 + 10.5 + 16) = 35$$

$$\frac{\partial L}{\partial w} = -\frac{2}{4} (35) = \boxed{-17.5}$$

$$\boxed{w} = w - \eta \frac{\partial L}{\partial w} = 0 - 0.05 (-17.5) = \boxed{0.875}^*$$

$$\frac{\partial L}{\partial b} = -\frac{2}{n} \sum_{i=1}^n (y_i - y_i')$$

$y_i^{(0)} = 0$

$\rightarrow [2.5, 3, 3.5, 4]$

$$\sum [2.5 + 3 + 3.5 + 4] = \boxed{13}$$

$$\frac{\partial L}{\partial b} = -\frac{2}{4} (13) = \boxed{-6.5}$$

$$b = b - \eta \frac{\partial L}{\partial b} = 0 - 0.05 (13) = \boxed{0.325}^*$$

$$\boxed{y' = w_1 x + 0.325}^*$$

$$\boxed{\varepsilon_{CM}}$$

$$w = 0.875 \quad b = 0.325$$

$$y_i^{(1)} = 0.875x + 0.325 \quad x = 1-4$$

$$\frac{\partial L}{\partial w} = -\frac{2}{n} \sum_{i=1}^n (y_i - y_i') x_i$$

$$y_i^{(1)} = 0.875(1) + 0.325 \quad (3)$$

$$= 0.875(2) + 0.325 \quad (4)$$

$$\checkmark y_i^{(1)} = [1.2, 2.075, 2.95, 3.825]$$

$$\checkmark y_i = [2.5, 3, 3.5, 4]$$

$$(y_i - y_i') = [2.5 - 1.2, 3 - 2.075, 3.5 - 2.95, 4 - 3.825]$$

$$\rightarrow (y_i - y_i') = [1.3, 0.925, 0.55, 0.175] \checkmark$$

$$(y_i - y_i') x_i = [1.3(1), 0.925(2), 0.55(3), 0.175(4)]$$

$$(y_i - y_i') x_i = [1.3, 1.85, 1.65, 0.7]$$

$$\rightarrow \sum (1.3 + 1.85 + 1.65 + 0.7) = 5.5$$

$$\frac{\partial L}{\partial w} = -\frac{2}{4} (5.5) = -2.75$$

$$w = 0.875 - 0.05 (-2.75) = 1.0125$$

$$\frac{\partial L}{\partial b} = -\frac{2}{n} \sum_{i=1}^n (y_i - y_i')$$

$$[1.3, 0.925, 0.55, 0.175]$$

$$\sum (1.3 + 0.925 + 0.55 + 0.175) = 2.95$$

$$\frac{\partial L}{\partial b} = -\frac{2}{4} (2.95) = -1.475$$

$$b = 0.325 - 0.05 (-1.475) = 0.39875$$

$$\rightarrow y' = 1.0125x + 0.39875$$

ECM ✓

x	L_1 $y_i - y_i'$	L_2 $(y_i - y_i')^2$
1	1.3	1.69
2	0.925	0.8556
3	0.55	0.3025
4	0.175	0.0306
		+

$$L_2 = 2.87875$$

$$ECM = \frac{1}{4} (2.87875)$$

$$ECM = 0.71969$$